

Review

Global Regulations for Sustainable Battery Recycling: Challenges and Opportunities

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Abstract: With the rapid expansion of transportation electrification worldwide, the demand for electric vehicles (EVs) has increased dramatically, creating new and sustainable growth opportunities for the global economy. However, as the most expensive component of EVs, lithium-ion batteries pose significant sustainability challenges due to raw material consumption and supply chain constraints, as well as the complexities of end-of-life battery disposal and recycling. To address these concerns, many countries are actively establishing regulations to promote sustainable pathways for battery reuse and recycling. Despite these efforts, existing battery recycling regulations remain often inefficient and vary significantly across different countries in legal enforcement, producer responsibility, waste classification, recycling targets, design standards, public engagement, and financial incentives, particularly given the complexities of the global supply chain and resource distribution within the battery industry. Understanding these regulatory differences and establishing a unified framework are therefore crucial to ensuring sustainable and efficient battery recycling. This review provides a comprehensive analysis of the necessity of establishing robust regulations for sustainable development of battery recycling industry. The evolution and refinement of battery recycling regulations are deeply reviewed to identifying persistent gaps and challenges in key countries. Furthermore, we discuss the challenges associated with regulatory enforcement and propose strategies for developing a more cohesive legislative framework to ensure the effective utilization of retired batteries.



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1. Introduction

The growing urgency to address climate change has promoted a global shift towards renewable energy and electrification, particularly in the transportation sector [1–3]. Electric vehicles (EVs) have become a key solution for lowering greenhouse gas emissions and achieving a sustainable future [4]. EV adoption has skyrocketed in recent years due to advancements in battery technology, increasing consumer awareness, and supportive governmental policies [5–7]. As illustrated in Figure 1, global EV sales have surpassed 15 million units with a strong trend projected to exponentially increase in most of countries and regions in the coming decades [8]. Government initiatives play a crucial role in accelerating this expansion. For instance, the United States aims for EVs to constitute 50% of new vehicle sales by 2030. The European Union (EU) has gone further, requiring that every new car sold after 2035 must be a zero-emission vehicle. Similarly, China, the world's largest EV market, has set a target for EVs to make up 40% of all new car sales by the

year 2030. However, this rapid growth in EV demand also poses a critical sustainability challenge for emerging industries, particularly lithium-ion batteries (LIBs), the cornerstone of EV technology [9–11]. The surge in global battery demand has raised a range of issues that threaten the long-term sustainability of this sector. These include mounting pressure on the supply chains for critical raw materials, which are not only geographically concentrated but often extracted under environmentally damaging and ethical issues. Compounding these concerns is the impending wave of end-of-life (EoL) batteries. If not properly collected, repurposed, or recycled, these EoL batteries could result in serious environmental pollution, the loss of valuable materials, and heightened safety risks. Therefore, the establishment of effective, large-scale regulatory frameworks for sustainable battery recycling has become an urgent global priority. How such regulations are designed, implemented, and harmonized across nations to achieve the long-term sustainable development of this emerging industry forms the central focus of this review.

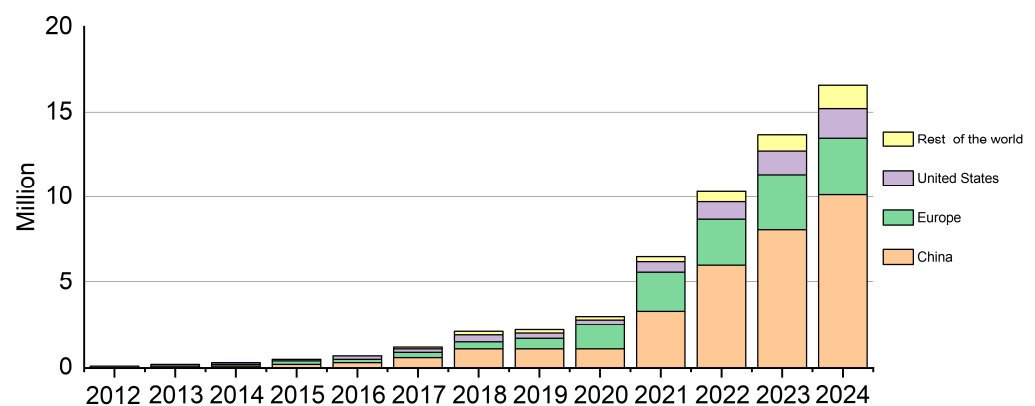


Figure 1. Global electric car sales during 2012–2024. Adapted from [8].

LIBs used in EVs typically last 8–15 years before reaching EoL in vehicles [12,13]. As EV adoption continues to accelerate, the world is on the cusp of a massive wave of battery retirements. Figure 2a shows that it is estimated that by 2030, LIBs’ recycling market will reach over 40 billion USD, with over 2 million metric tons of LIBs reaching EoL annually [14]. This trend is expected to rise further in subsequent decades, presenting significant environmental, economic, and logistical challenges, as shown in Figure 2b [15]. Studies have shown that improperly disposed LIBs can release heavy metals such as cobalt, nickel, and manganese, leading to soil and groundwater contamination exceeding safe regulatory thresholds [16–18]. Failing to recover these resources may worsen supply chain constraints and increase the cost of battery production by up to 30%, particularly for materials like lithium and cobalt, which are concentrated in politically unstable regions [19]. Therefore, integrating battery recycling into global sustainability strategies is essential. Recycling not only reduces environmental hazards but also enables the recovery of up to 95% of valuable metals such as cobalt and nickel, significantly decreasing dependence on primary extraction and improving supply chain resilience [20,21].

Beyond environmental concerns, economic and industrial factors are key drivers of sustainable battery recycling [15,22,23]. As shown in Figure 3a, the materials required for LIBs production, including cobalt, lithium, and nickel, are finite and concentrated in a few regions [24]. The growing demand for LIBs, coupled with the limited availability of raw materials, has led to volatility in the prices of these critical elements [19,25] (Figure 3b). Recycling can provide a stable, secondary supply of these materials, reducing reliance on mining and mitigating price fluctuations. Additionally, investment in battery recycling infrastructure can create new economic opportunities, fostering job creation in the recycling and materials recovery sectors. The transition to a circular battery economy can

also enhance the resilience of domestic supply chains, reducing exposure to geopolitical risks [26,27]. Furthermore, advancements in recycling technologies, such as direct cathode regeneration and hydrometallurgical processing, are improving the efficiency and cost-effectiveness of material recovery, making recycling an increasingly viable alternative to primary extraction [28–30].

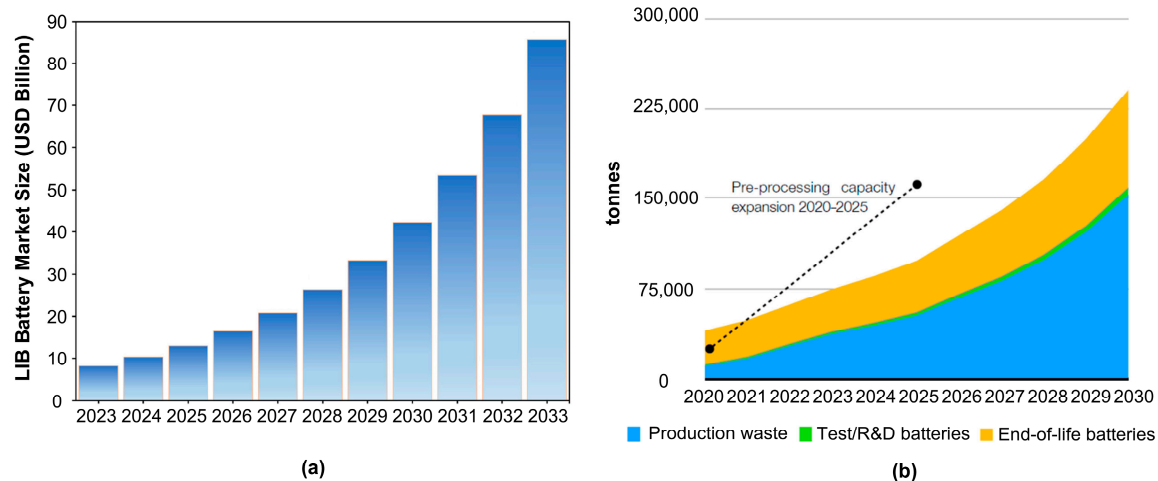


Figure 2. (a) LIB recycling market size during 2023–2033 (USD billion). Reproduced with permission from [14]. (b) Total volume of LIBs available for recycling in the United States (courtesy of Circular Energy Storage). Reproduced with permission from [15].

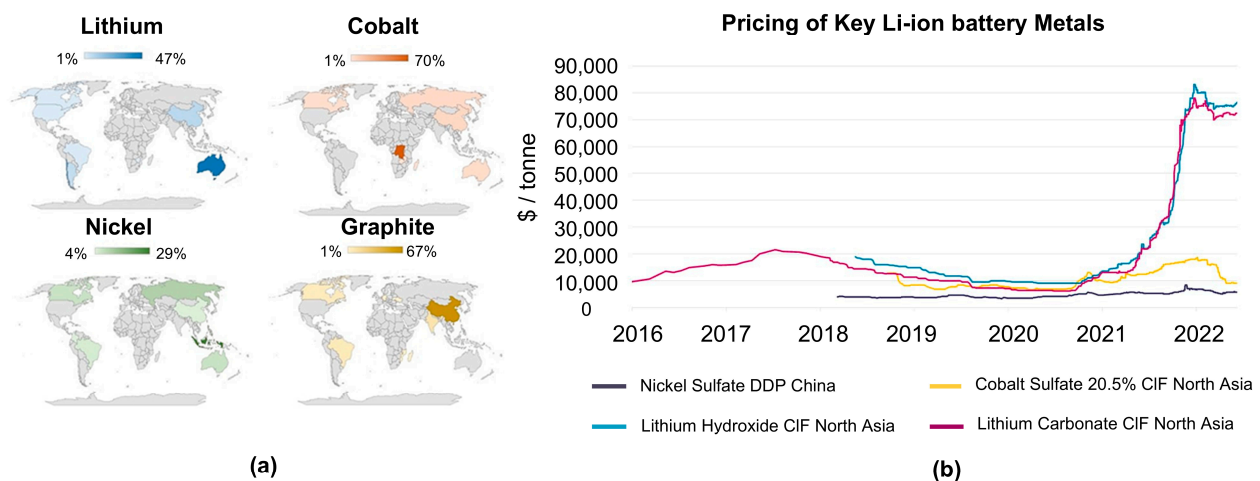


Figure 3. (a) Geographical distribution of lithium, cobalt, nickel, and graphite production. Reproduced with permission from [24]. (b) Price history of battery-grade key elements. Reproduced with permission from [25].

Despite these metrics, battery recycling remains in its early stages with relatively low efficiency [31]. To improve the recycling efficiency, many countries and regions have introduced new policies and regulations. This reflects a growing global commitment to sustainable resource management and environmental protection [32]. These frameworks are critical to institute standardized processes, ensure compliance, and incentivize investments in recycling facilities. While these efforts demonstrate encouraging momentum, the global development of battery recycling remains highly uneven. Different countries and regions face varying degrees of urgency and readiness for recycling, depending on the maturity of their battery industries, the volume of the EoL batteries, and domestic demand for critical materials. Regulatory frameworks also differ widely in scope, enforcement strength, and legal structure, which directly affects the sustainability and scalability of

recycling practices. Differences in institutional capacity, public awareness, and industrial infrastructure further complicate the global landscape. Given these disparities, analyzing region-specific developments is crucial for identifying best practices, systemic gaps, and policy innovation opportunities, especially understanding geographic regulatory frameworks and strategic priorities across key global players is essential for informing the design of globally coordinated, efficient, and sustainable battery recycling systems. For example, in the United States, battery recycling is promoted by federal programs and mandates by the states. The federal initiative is complemented by industry engagement through programs like the Battery Recycling Prize that incentivizes innovative recycling of LIBs. The EU published the Battery Regulation in 2023, which is the world's first comprehensive regulatory framework for batteries. This regulation sets strict rules for battery collection, recycling, and material recovery. It also requires manufacturers to publish information on material content and carbon footprint using digital battery passports, enhancing traceability and transparency. The regulation is expected to make Europe a global leader in sustainable battery management throughout its lifecycle. Being the world's leading market for EVs, China has enacted robust measures to manage EoL batteries and promote closed-loop battery recycling to achieve resource autonomy and sustainability. South Korea and Japan are also key players in battery manufacturing and have developed recycling frameworks. South Korea's Resource Circulation Policy requires recovery of key components from EoL batteries. Companies exceeding recovery targets receive financial rewards. Japan also supports battery recycling initiatives and often encourages the collaboration with industry partners. Automated disassembly systems are developed to improve the safety and efficiency of LIB recycling in Japan. India has introduced draft guidelines for battery recycling under its Battery Waste Management Rules in 2022, aiming to establish a circular economy for LIBs. Australia's Battery Stewardship Scheme focuses on voluntary Extended Producer Responsibility (EPR) programs to encourage responsible battery disposal and recycling.

Although the above efforts, establishing global harmonization remains very challenging [33,34]. The lack of standardized global regulations makes it difficult for manufacturers and recyclers to develop uniform strategies, resulting in inconsistencies in material recovery rates and the quality of recycled products. Furthermore, the technical and economic feasibility of scaling up recycling infrastructure remains a challenge, particularly in regions with underdeveloped waste management systems [35,36]. However, these challenges also present opportunities. Emerging technologies, including direct recycling and AI-driven material separation, have the potential to significantly improve the efficiency and economics of battery recycling [37,38]. Furthermore, international collaboration, supported by organizations such as the United Nations Environment Programme and the International Energy Agency, could help establish consistent recycling standards and facilitate cross-border material flows. Developing global agreements on battery recycling targets and best practices would ensure a more coordinated effort in achieving a circular economy for LIBs. As EV adoption continues to grow, addressing these challenges through innovation, policy harmonization, and public-private partnerships will be crucial for developing a sustainable and resilient battery ecosystem.

2. Sustainable Benefits of Establishing Battery Recycling Policies and Regulations

As the global concerns over retired batteries continues to rise, the establishment of robust battery recycling regulations has become a critical priority. Recycling regulations can lessen environmental hazards, contribute to a circular economy, reduce reliance on primary resource dependency, and advance climate objectives. This section explores the sustain-

able benefits of instituting battery recycling regulations, analyzing their environmental, economic, technological, and social impacts (Figure 4).



Figure 4. Sustainable benefits of establishing battery recycling regulations in environmental, economic, technological, and social impacts.

2.1. Environmental Sustainability

Battery recycling regulations are fundamental in minimizing environmental pollution by ensuring the proper handling, recovery, and disposal of hazardous battery components [39–41]. As shown in Figure 5, spent batteries contain toxic and flammable electrolytes like LiClO_4 , LiBF_4 , and LiPF_6 . They also contain heavy metals such as nickel, cobalt, manganese, and lithium. If not properly discarded, these materials can leach into soil and water, posing significant threats to ecosystems and human health [42,43]. Regulatory frameworks can mandate the safe collection, transportation, and treatment of battery waste, which prevents toxic leachates and air pollution from incineration [16,44]. Furthermore, recycling regulations could conserve natural resources by reducing reliance on primary mining. These mining processes are often associated with substantial greenhouse gas emissions, deforestation, habitat destruction, and excessive water consumption [45–51]. Recent lifecycle assessments have shown that recycling lithium-ion batteries into battery-grade materials can achieve substantial environmental benefits and reduce greenhouse gas emissions between 58% and 81%, contributing to climate change mitigation by reducing the carbon footprint of the battery industry. Furthermore, battery recycling requires from 77% to 89% less energy than raw material processing and from 72% to 88% lower water usage compared to conventional mining-based production processes [52]. Moreover, by recovering valuable metals such as lithium, cobalt, and nickel from spent batteries, recycling can mitigate critical supply constraints, reduce the ecological footprint associated with mining activities, and support more resilient supply chains. These environmental benefits align with the United Nations’ Sustainable Development Goals, particularly Goal 12 on responsible consumption and production and Goal 13 on climate action [53,54].

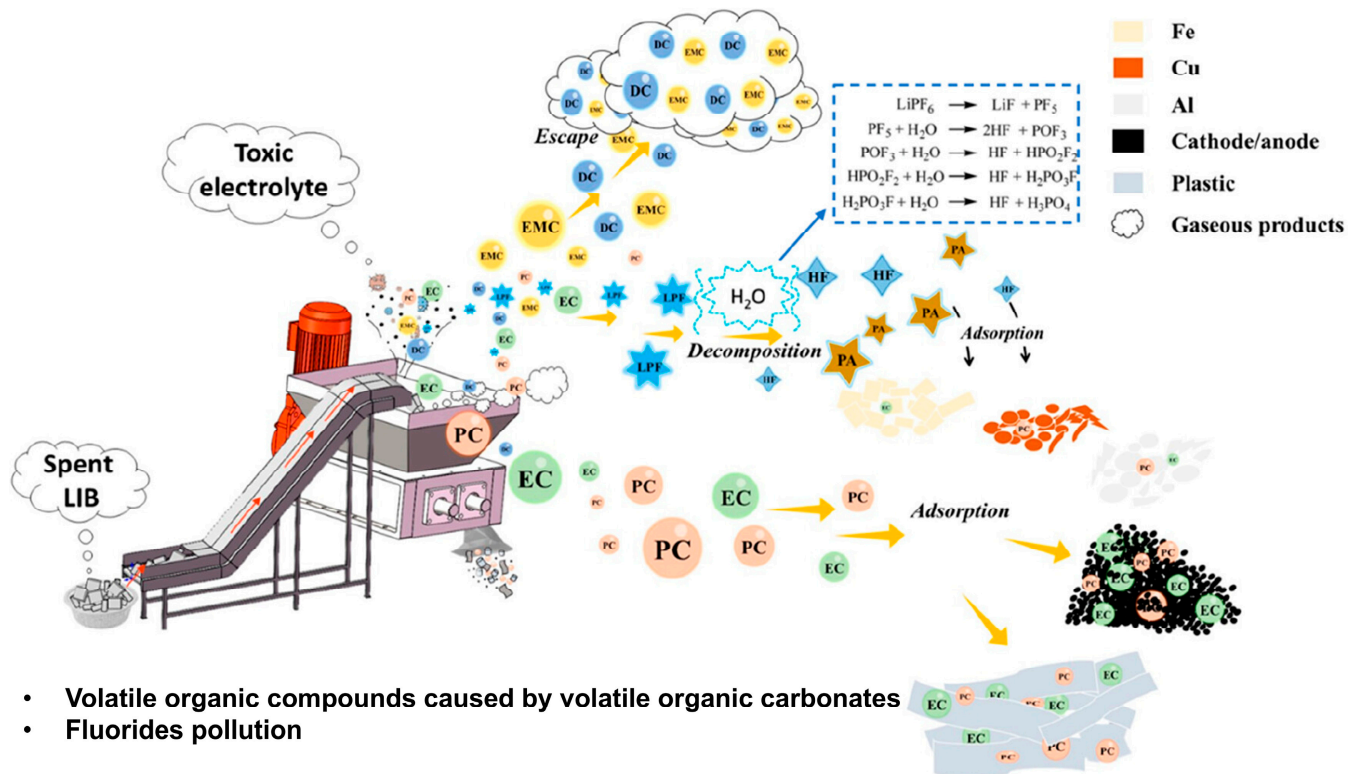


Figure 5. Schematic diagram of migration and transformation mechanism of hazardous pollutants throughout the recycling process. Reproduced with permission from [42].

2.2. Economic Sustainability

The implementation of battery recycling policies strengthens economic sustainability by fostering a circular economy, reducing reliance on scarce raw materials, and creating new market opportunities [55]. As shown in Figure 6, a circular economy framework for batteries emphasizes reusing, remanufacturing, and recycling materials, thereby reducing dependency on virgin resources and mitigating supply chain vulnerabilities [56–59]. This approach is particularly crucial given the geopolitical risks and price volatility associated with critical battery minerals, which are concentrated in a few countries as discussed above. By securing a secondary supply of materials through recycling, economies become more resilient to supply disruptions and market fluctuations. Recycling also lowers battery production costs, making EVs and energy storage systems more affordable and accelerating the transition to sustainable energy solutions [17,60,61]. The economic benefits extend beyond cost reduction to job creation and industry growth [62,63]. The expansion of the recycling industry generates employment opportunities in battery collection, transportation, material recovery, and research on advanced recycling techniques. Governments worldwide have recognized this potential, with initiatives such as the EU’s Battery Regulation [64,65]. As regulatory frameworks evolve, investment in battery recycling technologies will continue to drive economic growth while positioning nations as global leaders in sustainable energy solutions. Recycling policies also help keep material costs stable. This builds long-term investor confidence and supports reliable supply chains for manufacturers, energy providers, and consumers.

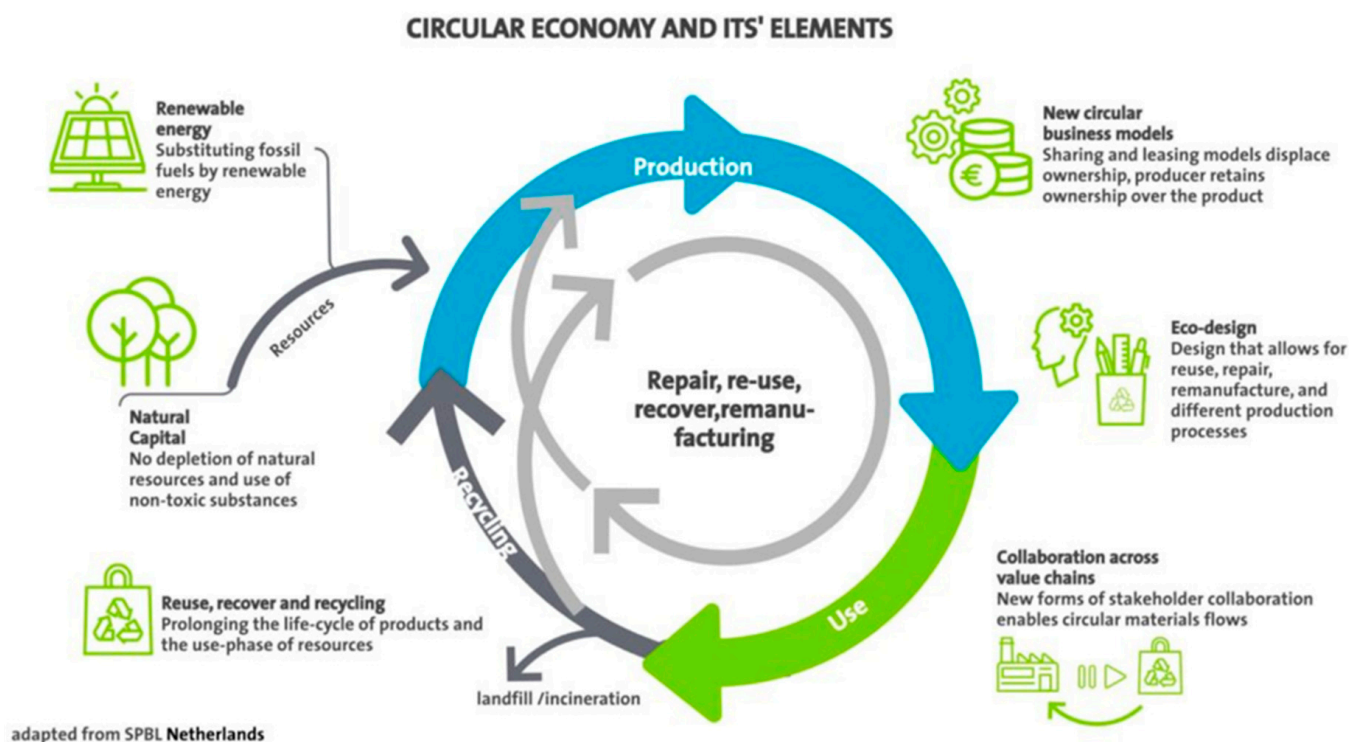


Figure 6. Circular economy strengthened by battery recycling. Reproduced with permission from [56].

2.3. Technological Advancements

Recycling regulations play a significant role to stimulate technological innovation in the battery industry to boost recycling efficiency [66,67]. Traditional battery recycling methods, such as pyrometallurgy, are characterized by substantial energy consumption and high emissions, which pose challenges to their long-term sustainability [68]. However, facilitated by regulations and policy initiatives, more environmentally sustainable recycling technologies, such as hydrometallurgy and direct recycling, have emerged. Hydrometallurgical processes offer higher recovery rates and lower environmental impacts [69]. Meanwhile, direct recycling, which preserves cathode materials for direct reuse in new batteries, further enhances resource efficiency and reduces the need for extensive reprocessing [70]. These technological advancements contribute significantly to enhancing the circularity of battery materials. Additionally, regulations that emphasize recyclability motivate battery manufacturers to reconsider battery design, incorporating modular designs, standardized chemistries, and easily separable components. This shift leads to longer lasting batteries, simplifies EoL recycling, and augments material recovery [70,71]. As governments and industries collaborate on recycling targets, these regulations not only advance sustainability but also drive the creation of new technologies for future energy storage applications.

2.4. Social and Ethical Sustainability

Implementing battery recycling rules has significant social and ethical value, especially in maintaining human rights violations associated with raw material mining. For example, the mining of cobalt has been associated with many ethical issues, including the exploitation of child labor, hazardous working environments, and great ecological damage. These issues pose significant risks, especially in the Democratic Republic of Congo [72,73]. Battery recycling regulations encourage the use of recycled materials to decrease dependence on ethically problematic mining operations. This will support fair labor standards and minimize human exploitation across global supply chains. Despite their intended goals, enforcing these regulation remains a significant challenge. While regu-

lations require manufacturers to document sourcing practices and comply with strict ethical procurement standards, inconsistent oversight can limit their effectiveness in promoting true transparency across the industry [18,74]. Many policies include consumer engagement tools, such as manufacturer-led take-back programs and deposit–refund systems, but their impact is frequently limited by low participation rates and a lack of monitoring or follow-through [75]. Strengthening enforcement and accountability measures is therefore essential to fully realize the benefits of these initiatives. These initiatives encourage people to return EoL batteries to designated locations so batteries can be recycled in an environmentally sound manner.

2.5. Alignment with Global Sustainability Goals

Battery recycling regulations are crucial to build clean energy systems and lower greenhouse gas emissions, eventually achieving worldwide climate and sustainability objectives. These regulations foster the efficient reuse of materials [76] and directly support the emissions reduction goals outlined in the Paris Agreement, whose aim is to keep the global temperature increase well below 2 °C. Moreover, recycling regulations could promote international cooperation to standardize best practices and harmonize regulatory frameworks [77]. For example, the EU’s battery regulation, which serves as a model for other regions, encourages global coordination on sustainable battery management. Through technological innovation and policymaking experience exchange, governments can collectively advance recycling efficiencies, reduce environmental footprints, and make global battery supplies more resilient [78].

Developing battery recycling regulations may improve environmental protection, boost economic growth, create technological innovation, and create social equity. However, obtaining these gains widely involves a holistic understanding of how numerous states formulate and implement battery recycling techniques. Regulatory frameworks differ substantially across regions, influenced by economic objectives, technological capacities, and existing legal structures [79]. While some nations have adopted strict recycling regulations, supported by strong infrastructure and enforcement measures, others remain in the preliminary stages of effective battery waste management. These disparities pose obstacles to harmonizing global recycling standards but simultaneously create opportunities for collaboration, shared learning, and policy innovation. As the battery industry continues to expand, bridging regulatory gaps across countries and establishing more sustainable and efficient worldwide recycling framework become increasingly important. To understand global recycling efforts, we must examine how countries create and apply their regulations, the challenges they face, and how they aim to improve sustainability. The next section explores the regulations of leading global actors, providing insight into their respective battery recycling strategies.

3. Regulation Frameworks Establishments and Developments in Key Global Players

3.1. China Policies and Regulations for Battery Recycling

China has gradually established a coherent regulatory framework for battery recycling [80]. Initially, regulations were scattered across various documents with limited coordination. However, with the rapid growth of EV industry, the government has enacted policies aimed at boosting recycling efficiency, advancing sustainability, and ensuring regulatory compliance. These efforts have culminated in a robust framework that enforces EPR, promotes traceability, and ensures regulatory oversight [81,82]. The following sections detail the historical evolution of China’s battery recycling regulations, highlighting key legislative and policy advancements.

In the early stage of the 1990s, regulations on EV power battery recycling were initially introduced, primarily as individual clauses within broader legislative frameworks. However, the content remained fragmented and lacked a systematic structure. The first formal regulatory framework about battery recycling emerged in 2003 when the State Environmental Protection Administration and other ministries issued the Technical Guidelines for Pollution Prevention and Control of Waste Batteries. LIBs were identified as one of the key targets for waste battery collection. It also explicitly requires battery manufacturers, importers, and manufacturers of products using batteries to take responsibility for the recycling of waste batteries. Additionally, it mandates relevant battery sellers to establish waste battery recycling facilities at their sales locations. These technical guidelines laid the foundation for the managing power battery recycling in China by clarifying responsible parties and facilitating the development of a structured recycling system. It also provided guidance for the development of subsequent policies. In 2007, China's "National 11th Five-Year Plan for Environmental Protection" explicitly proposed, for the first time, the establishment of an EPR system and the improvement of the renewable resource recycling system. In 2011, China's "National 12th Five-Year Plan for Environmental Protection" proposed advancing the implementation of the EPR system and promoting the efficient utilization of resources. In 2014, the State Council issued the Guiding Opinions on Accelerating the Promotion and Application of New Energy Vehicles, which suggested promoting the recycling of spent power batteries through methods such as funds, deposits, and mandatory recycling, and recommended studying and formulating policies for power battery recycling. At this stage, power battery recycling was only included as part of the provisions in policy documents related to the promotion and application of new energy vehicles. However, with the increasing popularity of new EVs and the growing usage of power batteries, the importance of power battery recycling has continued to rise. A series of policy documents laid the foundation for the advancement and formulation of subsequent regulations on power battery recycling. At the legal level, China's Solid Waste Law (1995) established the principles of "reduction, resource recovery, and harmless disposal", introducing cradle-to-grave lifecycle management. It also identified product producers, sellers, and users as the primary responsible parties. Though initially conceptual, this law laid the foundation for implementing EPR in EV battery recycling. The revised 2004 Solid Waste Law further clarified that any entity causing pollution must assume responsibility, making EPR more systematic and actionable. The 2008 Circular Economy Promotion Law reinforced the EPR system by explicitly incorporating reuse practices, providing stronger operational guidance for battery recycling. Collectively, these legal frameworks form the regulatory foundation for China's standardized battery recycling system within its broader circular economy legislation.

Beginning in 2015, widely considered as a key development phase of legislation, the government transitioned from policy guidance to practical implementation. The introduction of specialized policies for power battery recycling facilitated practical exploration of a recycling system. Pilot projects were launched to test recycling mechanisms, establish service outlets, and develop regionalized recycling networks. In 2016, Technical Policy on Electric Vehicle Power Battery Recycling (2015 Version) proposed establishing a battery coding system to ensure the traceability of EV power batteries. Power battery and EV manufacturers are designated as the main entities responsible for battery recycling. They are required to build recycling networks and designate at least one collection point in each city to handle EoL batteries. Joint efforts among multiple enterprises to build and share recycling networks were encouraged to reduce costs and improve efficiency. Additionally, measures such as deposits, buybacks, and trade-ins are proposed to incentivize consumers to return used power batteries. As a department rule and guiding document, this technical

policy focused on establishing technical standards and specifications for battery recycling. However, due to the lack of punitive provisions, enterprises faced insufficient constraints in fulfilling their recycling obligations, resulting in certain limitations in advancing the battery recycling system. The Plan for the Implementation of the EPR System, also issued in 2016, recommended that EV and power battery manufacturers utilize after-sales service networks to recycle EoL batteries, and Shenzhen has been designated as an initial pilot city for establishing the EV battery recycling and utilization system. A third-party credit certification and evaluation system was established, enabling third-party institutions to assess and verify enterprises' compliance with their EPR obligations. The above policies provided both policy support and practical pathways for establishing a power battery recycling system. By clarifying corporate responsibilities, implementing pilot projects, and introducing third-party supervision, these measures have gradually promoted the standardization and systematization of power battery recycling and utilization.

A major milestone came in 2018 with the Interim Measures for the Management of New Energy Vehicle Power Battery Recycling. This measure clarified, for the first time, that new EV manufacturers bear the primary responsibility for power battery recycling, while battery manufacturers and related recycling companies are tasked with coordination responsibilities. It also clarified the launch of the integrated management platform for national monitoring of new EVs and traceability of traction battery recycling. By utilizing battery coding as an information carrier, the platform ensures traceability of battery origins, defines responsibilities, and enables comprehensive lifecycle supervision of power batteries. The 2018 Notice of Seven Ministries on Effectively Advancing Pilot Work for the Recycling and Utilization of New Energy Vehicle Power Batteries decided to carry out pilot projects for the recycling and utilization of new EV power batteries in 17 regions, including Beijing-Tianjin-Hebei, Shanxi, Shanghai, Jiangsu, and China Tower Co., Ltd. These efforts laid the foundation for the nationwide promotion of power battery recycling and utilization. The 2018 Interim Regulations on Traceability Management of New Energy Vehicle Power Battery Recycling provide standardized guidelines for submitting relevant traceability information to the traceability management platform. The 2019 Guidelines for the Construction and Operation of Recycling Service Outlets for New Energy Vehicle Power Batteries requires automobile manufacturers to independently build, jointly establish, or authorize the creation of power battery recycling service outlets. These outlets must enhance information tracking and ensure accurate record-keeping throughout the disposal process of EoL batteries.

As power battery recycling entered a large-scale phase, the Chinese government accelerated policy development to establish a comprehensive regulatory framework. In 2022, power battery recycling was formally included in the Government Work Report, highlighting its strategic importance in national economic and environmental policies. Ongoing improvements to laws, regulations, and policy tools have laid a solid foundation for large-scale, standardized battery recycling in China. These efforts support a more sustainable and efficient circular economy. The 2021 Management Measures for the Cascading Utilization of New Energy Vehicle Batteries reinforces the EPR system and clarifying the path for producer responsibility. The 2023 Interim Measures for the Comprehensive Utilization of Power Batteries for New Energy Vehicles (Draft for Comments) is expected to become China's first legally binding regulation for power battery recycling. This regulation establishes a comprehensive lifecycle management system for battery recycling, ensuring traceability and accountability at every stage. In the design phase, power batteries must be assigned a unique identification code for full lifecycle tracking. During the installation phase, automobile manufacturers are required to record battery and vehicle code information. In the sales phase, manufacturers must inform consumers about battery recycling obligations and

procedures. The recycling phase clearly defines responsibilities: automobile manufacturers must recycle batteries installed in vehicles, while battery manufacturers are accountable for those directly sold to the market. Additionally, battery leasing operators, EoL vehicle dismantling enterprises, recycling service stations, and other relevant entities must fulfill their respective recycling responsibilities. Furthermore, the regulation clarifies the supervisory roles of national and local authorities, detailing regulatory requirements across all recycling stages to standardize and enhance the development of China's power battery recycling industry. Opinions of the General Office of the State Council on Accelerating the Construction of a Waste Recycling System 2024 emphasize strengthening the recycling of used power batteries and enhancing the traceability management of power batteries. It also proposes a joint inspection campaign targeting illegal "workshop-style" recyclers. By April 2024, this effort had shut down around 1800 small battery recyclers—about 20 closures per day. This increased regulatory oversight has driven the industry towards greater standardization and compliance. Comprehensive Utilization Standards for New Energy Vehicles Retired Power Batteries (2024 Edition) clearly defines specific requirements for the comprehensive utilization of EoL power batteries. It establishes detailed operational standards for key recycling processes such as disassembly, crushing, sorting, and smelting, aiming to maximize the recovery of valuable metals and materials, thereby promoting resource circularity and waste reduction. For comprehensive utilization enterprises, the document further specifies requirements for site selection, facility equipment, and traceability capabilities to ensure compliance with national environmental protection policies. As a guiding framework, this regulation enforces stricter standardization measures to enhance the effective recovery and rational utilization of power batteries, ultimately fostering resource conservation and environmental sustainability. At the legal level, China's revised Solid Waste Law (2020) mandates establishing a national solid waste management platform for comprehensive monitoring and digital traceability through big data. The law innovatively integrates solid waste management activities into the social credit system, motivating both corporate and individual compliance. It explicitly applies EPR to power batteries, requiring manufacturers to implement recycling systems proportional to their sales, independently or through authorized third parties. Enforcement has also intensified, empowering authorities to seize goods and imposing continuous daily fines until corrective actions occur. Additionally, dual penalties hold companies and executives jointly accountable, substantially raising the cost of non-compliance. These regulatory enhancements provide a robust legal foundation for China's standardized and sustainable battery recycling industry.

Despite these advancements, considerable challenges persist. As summarized in Figure 7, existing regulations governing power battery recycling and utilization are mainly issued by the Ministry of Industry and Information Technology, the State Council, and the Ministry of Science and Technology, in the form of departmental rules, industry standards, and technical guidelines. However, these regulations occupy a lower legal tier, are broadly framed, and remain fragmented, providing only general guidance without specific, mandatory, and enforceable implementation mechanisms. Additionally, ineffective EPR policies result in unclear accountability among manufacturers, sellers, and consumers. Insufficient public participation, due to a lack of awareness and economic incentives, further hinders progress. Strengthening legal frameworks, enforcement measures, industry regulations, and public engagement strategies is essential to improving recycling efficiency, fostering a circular economy.

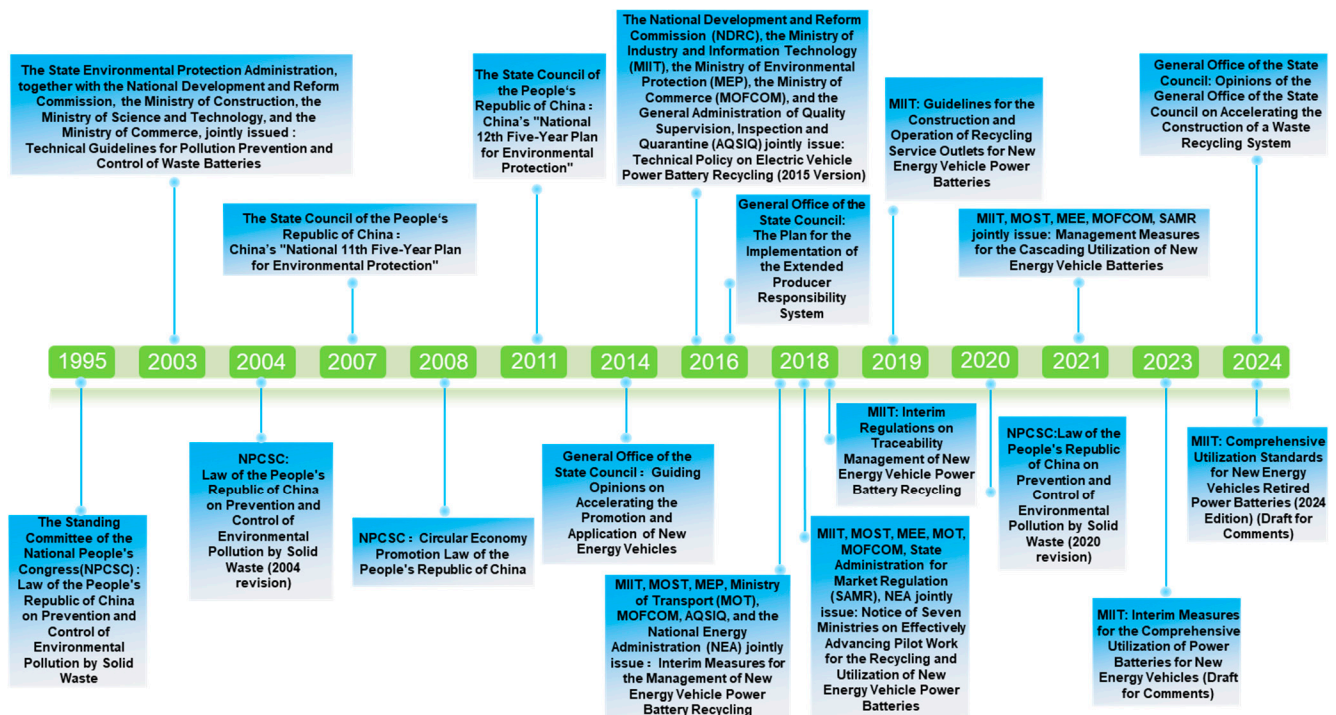


Figure 7. The timeline of battery recycling regulations in China.

3.2. European Union (EU) Policies and Regulations for Battery Recycling

The EU has taken a leading role in regulating EV battery recycling to promote a sustainable and circular economy, ensuring resource efficiency and reducing environmental impact [65,83]. Over the years, as summarized in Figure 8, the EU has progressively strengthened its policies to address the growing demand for LIBs, particularly in EVs, and the challenges associated with their EoL disposal.



Figure 8. The timeline of battery recycling regulations in the EU.

The EU Battery Directive 91/157/EEC was the EU's first regulation targeting batteries, aiming to reduce hazardous substances and mandate proper recycling. For more than a decade, it served as a blueprint for battery manufacturing and waste management, laying the groundwork for the EU's subsequent sustainable battery initiatives. The Waste Electrical and Electronic Equipment Directive 2012/19/EU sets collection and recycling targets for electronic devices containing batteries, indirectly influencing battery recycling efforts. End-of-Life Vehicle Directive 2000/53/EC requires vehicle manufacturers to assume

responsibility for the collection and recycling of EoL vehicle. This principle, referred to as EPR, ensures that manufacturers bear the financial and operational burden of proper disposal and recycling at the end of vehicles lifecycle. The directive stipulates that no more than 5% by the weight of an EoL vehicle should be sent to landfills by the year 2015. Producers must follow the International Organization for Standards guidelines for labeling and identifying vehicle components. While the directive does not directly regulate EV batteries, its main principles and requirements have helped the development of EV battery recycling policies and practices in the EU.

The EU Battery Directive 2006/66/EC and its 2013/56/EU amendment introduced EPR and established foundational requirements for battery collection, disposal, and recycling efficiency. In addition to these directives, the EU has launched initiatives to bolster EV battery recycling. In 2017, the European Battery Alliance was established to support each phase of the battery lifecycle, from raw material extraction and processing to manufacturing and recycling. One of its key goals is to ensure the EU's capacity to recycle and reintegrate valuable materials like lithium, cobalt, and nickel into the supply chain. The EU has also supported battery research and development, financing programs such as Horizon Europe, the Important Projects of Common European Interest, and the LIFE Programme, along with organizations such as the European Innovation Council and the European Investment Bank. These efforts drive innovation in recycling technologies, enhance material recovery, and promote collaboration among industry stakeholders, academia, and policymakers, ensuring more sustainable EV battery lifecycles.

Moreover, the European Green Deal (2019) and the Circular Economy Action Plan (2020) both called for stricter battery recycling rules to achieve long-term sustainability and reduce reliance on new raw materials. Then, the EU introduced the New Battery Regulation (EU) 2023/1542, which replaced the old directive, becomes a binding law that applies equally across member states. The regulation creates a comprehensive framework focused on sustainability, resource efficiency, and supply chain transparency. It also sets ambitious targets for battery collection and recycling to reduce environmental impact.

The Regulation's 96 Articles have been condensed into three main sections, and the core goal is enhancing the sustainability of spent EV batteries. The first section, representing the regulation's primary innovation, introduces collection and recycling targets, sets ambitious goals for battery recovery, material reuse, and the incorporation of recycled content. These measures significantly decrease dependence on newly mined raw materials and foster a circular battery economy. By 2027, the EU mandates a 63% collection rate for portable batteries, increasing to 73% by 2030. Additionally, EV and industrial battery manufacturers must establish take-back programs to facilitate the proper collection, recycling, and repurposing of EoL batteries. The regulation also sets rigorous recycling efficiency goals, requiring a minimum 65% lithium-ion battery recycling efficiency by 2025 and recycling efficiency goal by 2030. To boost material recovery, the regulation sets minimum recovery rates for key materials. Lithium recovery must reach 50% by 2027 and 80% by 2031. For cobalt, nickel, and copper, recovery must hit 90% by 2027 and 95% by 2031. Another major requirement is the mandatory use of recycled materials in new batteries. By 2031, new batteries must contain at least 16% recycled cobalt, 6% recycled lithium, and 6% recycled nickel. This will reduce the need for mining raw materials. The second section focuses on improving transparency and traceability throughout the EV battery lifecycle. It mandates the systematic collection and reporting of data on battery composition, performance, and environmental impacts through tools such as digital battery passports. The regulation further strengthens EPR by holding battery producers accountable for EoL batteries collection and recycling, and battery due diligence policies is established to ensure responsible sourcing and comprehensive supply chain assessments. The third section is

about lowering the climate impact during the battery's life. Manufacturers must share their carbon footprints. Maximum emission limits will be set in the future. These are designed to obtain high-quality data that supports environmental accountability and enables effective monitoring of compliance by industry stakeholders.

This regulation has a big effect on sustainability and battery rules around the world. It improves recycling and increases material recovery, so the EU depends less on mining. This helps protect the environment and reduces social problems from raw material extraction. It also lowers the carbon footprint of making batteries and supports the EU's climate goals in the Green Deal. Furthermore, by promoting a circular economy, where materials from retired batteries are systematically recovered and reused, the regulation limits hazardous waste and prevents pollution from improper disposal. This regulation sets an influential global standard, encouraging other regions, including the US, China, and other industrialized nations, to set higher sustainability standards and circularity in battery production.

3.3. *The US Regulations for Battery Recycling*

Battery recycling in the US is gaining significant momentum, driven by federal initiatives, state-level mandates, and the increasing urgency to establish a circular economy for critical minerals [22,84–86]. While no comprehensive federal law explicitly governs the recycling of EV batteries, several enacted laws, proposed legislation, and federal programs have shaped the regulatory framework. The Resource Conservation and Recovery Act of 1976 provides a fundamental “cradle-to-grave” regulatory system for tracking and regulating the generation, transportation, treatment, storage, and disposal of hazardous waste. It can apply to spent lithium-ion EV batteries if they are deemed hazardous due to their flammable electrolytes and reactive characteristics. Similarly, The Mercury-Containing and Rechargeable Battery Management Act (1996) primarily focused on lead–acid and nickel–cadmium batteries and mandated that battery manufacturers adopt conspicuous labeling and designs that facilitate disassembly for recycling. Lead–acid batteries, with unified technology and design, are comparatively easier to recycle extensively. In contrast, LIBs vary in materials, shapes, and sizes, complicating disassembly and causing fluctuations in component values that affect recycling profitability. Although this act does not specifically address EV batteries, it is often regarded as a potential template for EV battery recycling regulations. However, while existing lead–acid battery recycling legislation provides instructive insights for future LIBs policies, policymakers must recognize the significant technological and design differences between these battery types to develop effective legal frameworks for EV battery recycling.

More recent legislative efforts include the Battery and Critical Mineral Recycling Act of 2020 and 2021, which aimed to support the development of advanced battery recycling and the recovery of critical minerals. Although not passed as standalone bills, its provisions, such as support for battery recycling research and development, funding for pilot projects, and incentives to bolster domestic critical mineral supply chains, were later integrated into broader legislation, notably the Infrastructure Investment and Jobs Act (2021) and the Inflation Reduction Act (2022). The Infrastructure Investment and Jobs Act 2021 proposes the establishment of a ‘Battery Recycling and Second-Life Applications Program’ to support and award funds to the research, development, and application of advanced technologies aimed at increasing the recovery rate and purity of critical minerals in EV batteries. Additionally, this act convenes a task force to develop an extended battery producer responsibility framework, outlining the regulatory pathways for effective recycling. These will promote recycling EV batteries to reduce reliance on critical mineral

imports. Inflation Reduction Act (2022) offers tax incentives and grants to promote domestic EV battery recycling.

In addition to those federal laws, the US DOE has been a key driver in advancing battery recycling technologies [63]. Initiatives such as the ReCell Center (Figure 9), the Lithium-Ion Battery Recycling Prize, and funding commitments exceeding 192 million USD highlight a concerted effort to develop cost-effective and energy-efficient EV battery recycling solutions. Executive Order 13817 underlines the federal government's strategic priority to ensure a secure and stable supply of critical minerals by promoting battery recycling and reprocessing. Moreover, the National Blueprint for Lithium Batteries (2021–2030) envisions a closed-loop EV battery ecosystem, integrating research, infrastructure, and policy measures to enhance material recovery and reuse.

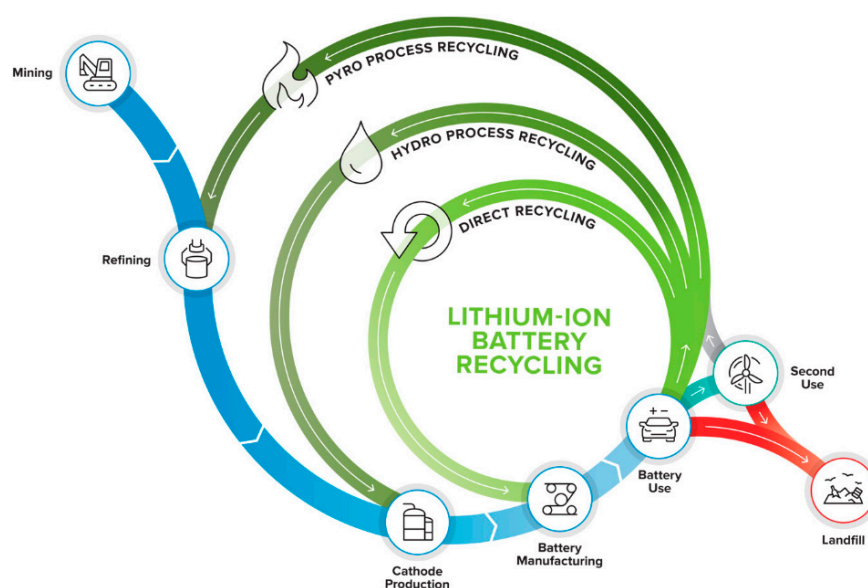


Figure 9. Lifecycle of a LIB proposed by the ReCell center of the US, including minerals to battery production and technical routes for battery. Image credit: Argonne National Laboratory.

At the state level, regulatory frameworks vary widely, reflecting diverse approaches to battery recycling. California has classified LIBs as universal waste, facilitating streamlines handling while enforcing strict disposal rules. The 2018 Assembly Bill 2832 established the Lithium-Ion Car Battery Recycling Advisory Group under the California Environmental Protection Agency (CalEPA) [87]. Following consultations with universities, manufacturers, and recycling stakeholders, the group proposed an EPR framework. This model holds manufacturers accountable for the proper recycling and disposal of batteries. The group also emphasized the need for standardized recycling systems and the expansion of second-life battery applications. It advocated for increased research and development investment to improve recycling technologies, as well as for stricter environmental and safety regulations for recycling facilities. Furthermore, public education initiatives are proposed to enhance awareness and encourage responsible battery disposal practices. These measures aim to achieve near 100% battery recycling, supporting California's sustainability goals and advancing a circular economy. North Carolina's House Bill 329 (2019) assigned the Environmental Management Commission and the Department of Environmental Quality to study and develop rules for reusing, recycling, and disposing of stationary energy storage system batteries [88]. The 2021 report of the agencies concluded that EV batteries composed of lithium-ion exhibit hazardous characteristics. Therefore, they concluded that the hazardous waste regulations in place are sufficient for such batteries, and additional regulatory programs are not necessary.

Other states have explored to integrate EV battery recycling into their environmental policies. In 2020, Arizona introduced a bill proposing the creation of an EV battery recycling fund, a fee on individuals selling or leasing EV batteries, mandatory processing at state-approved facilities, and a prohibition against landfill disposal; however, the bill ultimately failed to pass. Although the Arizona bill did not pass, it serves as a valuable example of a regulatory framework that could facilitate the reuse and recycling of large-format LIBs. Washington is expanding its E-Cycle program, originally designed for electronic waste, to consider a stewardship model for EV batteries and large-scale energy storage. Oregon, which has a strong history of product stewardship, is also considering including EV batteries in its broader environmental policies. These efforts often include grants and incentives to promote local recycling infrastructure and drive innovation. Several states have adopted innovative approaches. New York mandates in-store take-back for rechargeable batteries. Vermont launched the nation's first EPR program for single-use batteries. Colorado supports second-life applications through public–private partnerships—all aimed at improving EV battery sustainability. Meanwhile, Maine, Minnesota, and Texas are adapting existing e-waste recycling frameworks and piloting projects in collaboration with private sector stakeholders.

Despite these advancements, the US still lacks a unified national policy for EV battery recycling, leaving a fragmented regulatory landscape (Table 1). Establishing a clear and consistent federal, state, and local framework for LIB recycling could reduce regulatory uncertainty, mitigate liability concerns, and lower investor risks. While federal funding and research initiatives support technological advancements, the absence of a comprehensive, standardized legal framework poses challenges to scalability, efficiency, and investment in recycling infrastructure. Moving forward, integrating best practices from other policies on battery recycling, with more stringent EPR policies, outreach efforts, and industry collaborations, will be critical in developing a closed-loop EV battery material system for the US.

Table 1. The timeline of battery recycling regulations in the US.

Year	Regulations' Title	Federal/ States
2020	The Battery and Critical Mineral Recycling Act	Federal
2021	The Infrastructure Investment and Jobs Act	Federal
2021	The National Blueprint for Lithium Batteries (2021–2030)	Federal
2022	The Inflation Reduction Act	Federal
2023	The US Department of Energy announced a \$192 million funding opportunity for recycling batteries	Federal
2024	The US Department of Energy (DOE) funding commitments exceeding 192 million USD	Federal
2010	The Rechargeable Battery Law	New York
2014	H.695 (No. 139. An act relating to establishing a product stewardship program for primary batteries)	Vermont
2018	Assembly Bill 2832	California
2019	House Bill 329	North Carolina
2020	House Bill 2828	Arizona
2021	D.C. Law 23-211. Zero Waste Omnibus Amendment Act of 2020	Washington D.C.
2024	S.253 (No. 152. An act relating to including rechargeable batteries and battery-containing products under the State battery stewardship program)	Vermont

3.4. Other Regions Policies and Regulations for Battery Recycling

To address environmental concerns, resource scarcity, and rising battery waste, the other regions are also actively developing its battery recycling system [88]. Japan, as one of

the major players of battery manufacturing, have developed a sequence of legal frameworks, starting with the 1991 Act on the Promotion of Effective Utilization of Resources, laying the groundwork for EPR. The act emphasized the importance of incorporating recyclability into product design and ensuring proper EoL disposal. Later legislation, such as the Law for the Recycling of End-of-Life Vehicles of 2001, also made provisions for proper waste and hazardous material treatment. The 2013 Law for the Promotion of Sorted Collection and Recycling of Small Waste Electrical and Electronic Equipment extended recycling responsibilities to small rechargeable batteries, reflecting the increasing presence of portable electronics.

The principle of EPR, consumers' participation, hazardous substances management, and transparent recycling targets are established and developed in Japan's battery recycling regulations. Manufacturers and importers are required to properly recycle batteries, to design products for easier disassembly, and to fund recycling programs. Public campaigns educate consumers to send their EoL batteries to collection points, and this minimizes waste dumped into garbage and pollution of the environment. The Chemical Substances Control Law and the Public Cleansing and Waste Management Law regulates strict hazardous substance control to ensure safe treatment of hazardous chemicals such as cadmium, lead, and mercury. Manufacturers are required to meet recycling targets imposed by government. The Japanese government has also established clear reporting requirements for recycling progress, fostering transparency in the industry. Japan faces several challenges in managing EoL EV batteries. Unlike the EU, it does not have specific and comprehensive regulations governing EV battery recycling. Instead, the Ministry of the Environment and the Ministry of Economy, Trade, and Industry are the primary agencies responsible for regulating battery recycling. These two main ministries must collaborate to draft legislation and need approval from their Advisory Councils, which are made up of industry and academic representatives. A consensus from the councils is necessary for the draft to proceed to cabinet discussions and adoption. While this thorough process ensures careful deliberation, it also contributes to delays in establishing specific EV battery recycling laws. Japan's battery recycling efforts support sustainability both domestically and globally. Looking ahead, Japan plans to enact targeted regulations, expand consumer awareness, and work with international partners to establish standardized battery recycling guidelines.

Another representative is South Korea, whose regulatory framework focuses on recycling, sustainability, and resource efficiency throughout the EV battery lifecycle. In 2023, the Korean Agency for Technology and Standards launched a plan to standardize the recycling of EoL EV batteries, promoting collaboration among industry, academia, and research institutions. That same year, the Waste Management Act was revised to extend the storage period of EoL batteries from 30 to 180 days, offering recyclers greater flexibility. In 2024, EV batteries were classified as "recyclable resources". This classification exempts undamaged and repurposed batteries from stringent waste regulations. From 2027, EV batteries will undergo mandatory performance assessments before removal, determining whether they can be remanufactured, reused in energy storage systems, or processed for material recovery. Additionally, the specified percentage of recycled materials is required in newly produced EV batteries, reducing reliance on raw materials. However, some enforcement challenges need to be addressed in the future, such as the high costs of recycling infrastructure, difficulties in assessing battery health, and the need for greater industry-wide collaboration. The classification of EoL batteries as recyclable resources raises safety and quality concerns since not all repurposed batteries may meet performance standards. Effectively addressing these enforcement challenges is key for the long-term success of South Korea's battery recycling system.

Other nations also introduce regulations and policies to manage the EV battery life-cycle (Table 2). For example, although Canada lacks a comprehensive federal producer responsibility statute, provinces such as British Columbia, Quebec, Manitoba, and Ontario have implemented compulsory recycling directives for different types of batteries. These provincial programs typically incorporate EPR, which requires manufacturers and importers to oversee the collection and recycling of used batteries. Australia has recognized the importance and efficiency of battery recycling in its National Battery Strategy. The strategy develops recycling markets and infrastructure to handle EoL batteries responsibly. In India, the Battery Waste Management Rules of 2022 requires producers to collect and recycle retired batteries, which covers EV, portable, and industrial batteries. These rules set specific targets for material recovery efficiency and stipulate minimum percentages of recycled materials to be used in new EV and industrial batteries. Nigeria's National Environmental (Battery Control) Regulations of 2024 seek to mitigate battery-related pollution by integrating an EPR framework, forbid the import and export of used batteries without proper permits, and set forth guidelines for their collection, transportation, and storage. Collectively, these regulatory advancements reflect a global commitment to sustainable battery management, resource conservation, and the advancement of a circular economy in the EV sector.

Table 2. The timeline of battery recycling regulations in some countries.

Year	Regulation Title	Countries
1970	Waste Management and Public Cleansing Law	Japan
1973	Chemical Substances Control Law	Japan
1991	The Act on the Promotion of Effective Utilization of Resources	Japan
2002	Law for the Recycling of End-of-Life Vehicles	Japan
2013	Law for the Promotion of Sorted Collection and Recycling of Small Waste Electrical and Electronic Equipment	Japan
2022	Battery Waste Management Rules	India
2023	The Waste Management Act (2023 revision)	Korea
2024	The Act on Promotion of the Transition to Circular Society	Korea
2024	National Battery Strategy	Australia
2024	National Environmental (Battery Control) Regulations	Nigeria

4. Challenges in the Implementation of Battery Recycling Regulations

Despite growing global attention on battery recycling, the implementation of these battery recycling regulations faces many challenges [89]. Regulatory frameworks vary widely across countries, creating fragmentation that complicates compliance, hinders international cooperation, and allows regulatory loopholes [22,90]. Low public awareness and insufficient incentives limit participation in recycling programs, while high costs related to transportation, processing, and hazardous waste compliance challenge the economic viability of recycling. Technological barriers further complicate recycling efforts, as existing methods struggle with efficiency, environmental impact, and the growing diversity of battery chemistries. Addressing these challenges requires stronger regulatory coordination, increased public engagement, economic incentives, and continuous technological innovation. The following sections discuss the challenges facing the implementation of battery recycling regulation across different countries, highlighting the lessons they offer for developing a more harmonized and sustainable global framework (Figure 10).



Figure 10. Current challenges facing the implementation of battery recycling regulations.

4.1. Fragmented Regulatory Frameworks in EV Battery Recycling

Regulations governing EV battery recycling across major markets are notably fragmented, localized, and typically of lower legal status, causing significant challenges for implementation and enforcement [91]. For example, in the United States, EV battery recycling is regulated through a patchwork of federal guidelines and state specific regulations rather than a unified, comprehensive framework. This fragmented approach creates uncertainty, as companies must navigate varying compliance requirements depending on where the batteries are processed or sold. In China, most EV battery recycling management regulations are issued by the Ministry of Industry and Information Technology, the State Council, the Ministry of Science and Technology, typically in the form of departmental rules, industry standards, and technical specifications. These guidelines are largely advisory rather than legally mandatory, resulting in less predictable enforcement. In contrast, the EU has established a more comprehensive and stricter framework Regulation (EU) 2023/1542, which unifies aspects of battery collection, recycling efficiency, and EPR. However, even within the EU, differences in enforcement capabilities and national interpretations of the regulation can lead to inconsistencies, complicating efforts to streamline cross-border operations [92,93].

The lack of harmonized global regulations heightens existing challenges, resulting in competitive imbalances, regulatory loopholes, and potential environmental hazards. Companies may exploit weaker regulations in certain regions to reduce compliance costs, leading to an uneven playing field among recyclers and battery producers. Also, the lack of clear rules makes cross-border battery trade and logistics harder. For example, a Japanese exporter may find that batteries that are fine for recycling at home need more testing and certification to meet EU rules. These extra steps slow down processing, increase paperwork, and raise operational costs. As a result, battery shipments to the EU may be delayed, reducing the exporter's competitive edge.

International vehicle and battery makers face more complexity when they follow different rules because they must deal with different regulations, from the EU's strict ones to the more scattered or regional regulations in the United States, China, and Japan. This variety of regulations raises costs and forces companies to create many strategies to meet different demands. As the global EV market grows, creating a clear legal framework with strong authority and more international collaboration becomes more important.

EV battery recycling industry also needs favorable market-pull government policies to avoid a “chicken and egg” scenario. Without strong regulatory incentives such as subsidies, tax credits, or low interest loans, companies are reluctant to invest in advanced recycling technologies and expanding necessary infrastructure, causing underdeveloped collection systems and low recovery rate of spent EV batteries. The resulting scarcity of recyclable EV batteries makes recycling operations economically unviable, as companies struggle to achieve economies of scale, thereby increasing recycling costs and reducing its appeal. This scenario forms a negative feedback loop: inadequate recycling capacity yields a limited supply of batteries for recycling, and that limited supply, in turn, discourages further investment in necessary infrastructure. Breaking this cycle requires coordinated efforts and strong incentives. These should support both recycling capacity expansion and the collection of enough spent EV batteries to build a sustainable and profitable recycling system.

4.2. Low Public Awareness in EV Battery Recycling: Challenges and Global Efforts

Many countries have explicitly incorporated the principles of information disclosure and public participation into their environmental protection laws to enhance transparency and foster public awareness and active public involvement in environmental governance [94–96]. The public has the right to access environmental information, and only when they are fully informed about matters related to environmental protection can they effectively exercise their right to participate. The right to know is the foundation of the public’s right to participate.

This principle is also important in the area of EV battery recycling. The lack of public awareness and the inadequate dissemination of information concerning the importance of recycling EV batteries present significant barriers to the global development of EV battery recycling [96]. Many people are unaware of the environmental and health hazards posed by improperly disposed waste batteries, such as soil and water contamination or the release of toxic chemicals. This lack of awareness blocks the participation in recycling programs and leads to a “bystander effect”. Limited public participation restricts the amount of material available for recycling, thereby reducing recycling rates. Without enough volume, recycling operations cannot reach the scale needed for cost-efficient processing. This raises recycling costs and discourages further investment in recycling infrastructure and technology. As a result, environmental damage from improper battery disposal continues unchecked, further harming sustainability efforts and public health. Ultimately, the combined effects of low public awareness and the bystander effect create a self-reinforcing cycle that impedes the development of a robust and sustainable recycling ecosystem.

Some countries have slowly introduced policies and government-led public education programs to increase public awareness and involvement. For example, Germany has run nationwide awareness campaigns, like the Gemeinsames Rücknahmesystem Batterien (GRS Batteries). This program teaches people about recycling through media campaigns and school programs. In China, authorities used the internet and local events to show the economic and environmental benefits of recycling. The Lithium-Ion Car Battery Recycling Advisory Group in California highlighted the need for public education campaigns to raise public awareness. Despite these efforts, the insufficient strong economic incentives, such as tax refunds or the deposit–refund system, is the predominant constraint for the motivation of the populace. For the limitations to be overcome, the institutions and the governments need to make the awareness raising activities and education their prime agenda, emphasizing the benefits for the conservation of resources for the long term and the mitigation of the impact upon the environment. Otherwise, the shortfall in awareness

will continue being the predominant constraint for the establishment of the scalable and durable recycling infrastructure around the globe.

4.3. Economic and Safety Challenges in Recycling Lithium Batteries

Recycling LIBs is a highly complex and expensive process, driven by both the diverse composition of their components and the stringent safety and regulatory requirements necessitated by their hazardous nature [39,97]. The multifaceted challenges associated with LIB recycling, ranging from intricate material separation to costly transportation, treatment, and disposal, create substantial barriers to its economic feasibility [40,98].

The high cost of recycling LIBs is largely driven by the complexity and diversity of their components. Within the cells, the chemical compositions of active materials vary by manufacturer and application and may never standardize. For instance, while LiCoO_2 (LCO) is common in consumer electronics, automotive batteries often employ various combinations of Ni, Mn, Co, and Al to optimize performance and lower raw material costs, or even alternative cathode materials like LiFePO_4 (LFP). Additionally, manufacturers use different materials for the anode, such as graphite, silicon, or their composite, and these variations necessitate intricate separation processes during recycling. The problem is worsened by the design of battery packs. Some, like those in the Tesla Model S, can contain up to 5000 cells. These are integrated into modules with complex circuitry and thermal systems, making them hard to disassemble. As a result, recycling involves numerous labor-intensive and technologically demanding steps, making the overall cost significantly higher than that of extracting and refining virgin resources. This wide-ranging complexity stops investment in recycling facilities. It also poses a major challenge for the economic viability of LIB recycling.

Additionally, many EoL batteries are classified as hazardous wastes. This raises transportation, treatment, and disposal costs and requires more effort to meet strict rules [99–101]. The high flammability of LIBs makes them prone to overheating, ignition, and fires, especially in cases of short circuits, physical damage, or improper assembly. Additionally, the potential for thermal runaway, a chain reaction leading to a violent release of stored energy and flammable gas, can potentially trigger large scale thermal events. This danger requires strict safety controls during transit and highlights the need for rigorous transportation regulations. Consequently, LIBs are classified as hazardous materials under the US Department of Transportation's Hazardous Materials Regulations (49 CFR Parts 171–180) and must comply with all applicable requirements during transportation. These include proper packaging, labeling, placarding, documentation, and adherence to specified quantity and transit conditions to ensure safe handling and transit. Due to their hazardous nature, EV batteries require additional expenditures and efforts for transportation, treatment, and disposal, as well as for meeting strict regulatory standards. Transportation costs significantly drive up the overall expenses of EV battery recycling, and safe transportation alone accounts for 40–50% of the total recycling cost, according to various sources. Furthermore, operations such as vehicle dismantling and battery pack disassembly are especially costly due to the inherent safety risks associated with these partially manual procedures.

As a result, the recycling process involves multiple complex steps that collectively drive up costs, making it considerably more expensive than extracting and refining virgin materials, and posing a significant challenge for the economic viability of recycling LIBs.

4.4. Recycling Technology Challenge

Current EV battery recycling technologies are generally categorized into three categories: pyrometallurgy, hydrometallurgy, and direct recycling, as shown in Figure 11 [102–106]. Pyrometallurgy involves feeding dismantled battery modules into a high-temperature furnace

(typically around 1500 °C), along with fluxing agents such as limestone and silica [107]. Organic compounds like binders and electrolytes are combusted, while valuable metals such as copper, cobalt, nickel, and iron are reduced into an alloy that is later processed for recovery. Although pyrometallurgy can achieve high recovery rates for cobalt and nickel (typically over 90%), it causes significant losses of lithium, aluminum, graphite, and electrolytes, which are either emitted as off-gas or retained in slag. The method also has a high energy demand, generates substantial CO₂ emissions, and requires expensive gas treatment systems. These limitations are particularly pronounced when processing newer cathode chemistries like LiFePO₄ or manganese-rich materials, which contain low concentrations of high-value metals.

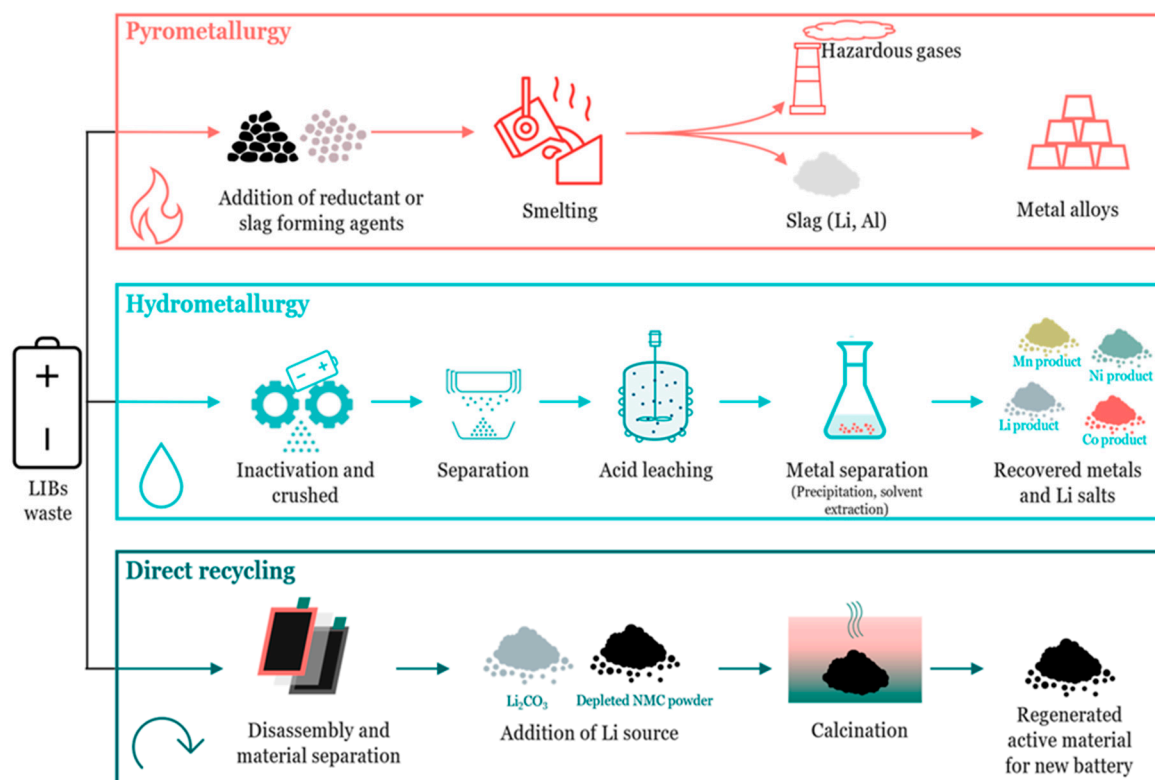


Figure 11. Three battery recycling routes, including pyrometallurgy, hydrometallurgy, and direct recycling. Reproduced with permission from [106].

In contrast, hydrometallurgy uses acidic leaching agents, such as H₂SO₄ with H₂O₂ or organic acids, to dissolve lithium and transition metals into solution, followed by selective precipitation or solvent extraction to recover Li, Co, Ni, and Mn [12,108]. This approach operates at much lower temperatures (typically under 100 °C) and offers high recovery efficiency, often exceeding 95% for key metals. However, it also produces large volumes of wastewater and chemical byproducts that require careful handling and treatment. The method's economic viability is reduced when recycling batteries with low cobalt or nickel content, such as LFP. While lithium recovery (e.g., as Li₂CO₃) is relatively mature, the recovery of electrolytes and organic additives remains underdeveloped. Recent advancements, including selective leaching, bioleaching, and electrochemical separations, are improving the environmental impact and selectivity of hydrometallurgical processes.

Direct recycling takes a fundamentally different approach by attempting to recover active battery materials, such as cathode powders, while preserving their structural integrity and electrochemical performance. Methods such as thermal relithiation, hydrothermal treatment, and chemical repair are used to restore the functional properties of degraded

materials without decomposing them into elemental forms [109,110]. This pathway offers significant environmental and economic advantages by reducing energy consumption and avoiding intensive chemical processing. It is estimated that direct recycling can retain 80–100% of the original performance of the cathode materials while saving up to 40% of energy compared to traditional routes. However, direct recycling is still at an early stage of development and faces challenges, including the lack of standardized battery designs, difficulties in sorting mixed chemistries, and maintaining consistent quality of regenerated materials. The US DOE's ReCell Center has demonstrated progress in direct regeneration of NMC cathodes and is developing AI-assisted sorting technologies to improve process adaptability and scalability.

Overall, while pyrometallurgy and hydrometallurgy remain the dominant industrial approaches for EV battery recycling, both face limitations related to energy intensity, environmental impact, or economic viability. Direct recycling, although still emerging, holds the greatest promise for achieving closed-loop recycling with minimal energy input, reduced emissions, and higher resource efficiency. Moving forward, advances in smart sorting, cathode rejuvenation, and unified battery design standards will be key to scaling up sustainable recycling technologies and transitioning from raw material recovery to true circular material regeneration.

5. Perspective and Outlook

As battery recycling becomes a global issue, international coordination is crucial for addressing the challenges posed by various regulations and measures of electric vehicle battery recycling across different countries. Without comprehensive and enforceable battery recycling regulations, the world risks facing serious environmental, economic, and social consequences. Improper disposal of spent batteries could lead to widespread contamination of soil and water by toxic metals and electrolytes, posing threats to ecosystems and public health. Economically, the loss of recoverable materials such as lithium, cobalt, and nickel would exacerbate raw material scarcity, increase production costs, and heighten supply chain vulnerabilities. Socially, continued reliance on primary resource extraction, often in regions with poor labor protections, could perpetuate human rights violations and geopolitical tensions. In the absence of regulation, fragmented and inefficient recycling systems would persist, undermining efforts to build a global circular economy and jeopardizing long-term climate and sustainability targets. Therefore, some key markets such as the EU, the US, Japan, and China should take lead to establish a coordinated, comprehensive approach for robust international collaboration for sustainable and efficient battery recycling. These unified set of regulations for all major producing and consuming countries could reduce uncertainty for multinational companies, simplify international logistics, and create fair competition for manufacturers and recyclers.

A critical step towards global coordination is the development of a harmonized digital “battery passport” system, which records each battery's composition, origin, usage history, environmental footprint, and real-time state-of-health (SoH). Artificial intelligence (AI) can enhance this system by analyzing data from battery management systems to predict degradation trends and remaining lifespan, enabling automated alerts for reuse, repurposing, or recycling. Integrating AI with battery passports would support dynamic, cross-border lifecycle management by manufacturers, recyclers, and regulators. However, this also raises concerns about data privacy and intellectual property. To ensure secure and ethical data sharing, the system must incorporate strong encryption, access controls, and anonymization protocols, backed by international regulatory alignment with privacy laws.

In addition to unified regulations, robust and forward-thinking public engagement is equally critical. An informed and motivated public can significantly increase partici-

pation in battery return, collection, and sorting programs, thereby boosting the supply of recyclable materials and reducing per-unit recycling costs through economies of scale. Governments and industry stakeholders should adopt innovative strategies beyond traditional awareness campaigns. For instance, a digital “battery recycling credit” system could reward consumers with discounts, tax credits, or loyalty points for returning used batteries, incentivizing responsible behavior. AI-enabled apps or in-vehicle dashboards could also nudge users with personalized reminders as batteries near end-of-life, guiding them to certified drop-off points and showing real-time environmental or economic benefits of recycling. By embedding such mechanisms into broader recycling policy, governments, manufacturers, recyclers, and consumers can together cultivate a circular economy for EV batteries. Moreover, this engagement can incentivize manufacturers to adopt “design for recycling” principles, simplifying disassembly and recovery. In the long term, it can also fuel the growth of industries centered on battery refurbishment and second-life applications, such as stationary energy storage. Together, these efforts will extend battery lifecycles and accelerate the broader transition to a resource-efficient, low-carbon energy system.

Technological innovation in battery recycling should be accelerated through international collaborative pilot projects. Currently, pyrometallurgical and hydrometallurgical recycling dominate but have significant drawbacks. Pyrometallurgy consumes high energy and loses critical materials like lithium and aluminum, reducing its environmental and economic viability. Hydrometallurgy, though less energy-intensive, uses extensive chemical reagents and generates problematic wastewater. Both methods struggle with efficiently recovering battery electrolytes. Given regional differences in battery supply chains, technological strengths, and recycling infrastructures, inter-national collaboration is essential. Countries with robust battery manufacturing, such as China and South Korea, can contribute automated collection and disassembly techniques, while regions like the EU and Japan can offer advanced hydrometallurgical re-cycling expertise. Nations rich in renewable energy, such as the US and Canada, could pilot low-carbon recycling technologies. Such collaborative projects, involving governments, industry, and academia, could validate advanced closed-loop recycling methods (particularly direct recycling), develop international standards for EV-grade materials, and explore scalable economic frameworks like cross-border cost-sharing for collection and processing. Existing models, such as the ReCell Center in the US, highlight how targeted funding and international partnerships drive technological breakthroughs. Expanding these cooperative efforts globally would greatly improve battery recycling efficiency, reduce environmental impacts, and advance the global circular economy for electric vehicle batteries.

Market-based benefits have been seen as a strategy to accelerate the adoption of sustainable recycling practices. Various financial incentives, such as tax rebates, feed-in tariffs for secondary raw materials, deposit-refund schemes for battery purchases, or government grants for building modern recycling facilities, can align market forces with environmental goals. Another financial challenge facing efficient recycling is the high cost of the transportation of spent batteries. Facilitating regional integration through international agreements and harmonizing regulations could substantially reduce these costs by streamlining logistics, lowering compliance costs, and enhancing recycling efficiency and safety. Along with stricter enforcement and strong reporting rules, these measures push companies to incorporate recycling into the entire lifecycle of batteries, from design and production to collection and disposal. In addition, increasing minimum recycled content rules for new batteries, as some laws have proposed, can establish a steady demand for recycled materials, thereby increasing market confidence and driving private investment in recycling technologies. If implemented globally, these actions could significantly reduce

the industry's reliance on virgin raw materials, ease supply risks, and mitigate the social and environmental impacts associated with mining.

In summary, advancing EV battery recycling requires a multifaceted strategy that addresses fragmented regulation, leverages public engagement, and spurs technological innovation. Strong international collaboration can unify disparate regulations, while heightened consumer awareness can boost collection rates and lower overall recycling costs. Simultaneously, continued investment in research can accelerate breakthroughs in direct recycling and refine existing processes, making them more cost-effective and environmentally sound. Taken together, these measures will be critical for transforming battery recycling into a sustainable, profitable component of the broader circular economy for energy storage. By aligning policy, technology, and community engagement, stakeholders can minimize environmental harm, conserve valuable resources, and ensure the long-term viability of electric vehicle adoption as part of a more comprehensive transition to sustainable mobility.

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